SP
BP
Si
Di
offset

1 SS
SS
DS
SS

Question #1: First Question (10 marks): Fill-in the table with the correct answer:

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
c	A	B	B	B	B	c	c	c	D

10

1. Address bus width of Intel 8086 microprocessor is:
   A. 16 B. 24 [✓ C. 20] D. 32
2. The physical address of the source operand in the instruction MOV AX, [BX]. Assuming DS register= 2000H, and BX register= 150H, is:
   [✓ A. 20150H] B. 2150 C. 20000 D. 200150
3. The segment that contains the assembly language instructions to be executed by the microprocessor is
   A. Data segment [✓ B. Code segment] C. Stack segment D. Extra segment
4. Instruction Pointer (IP) contains offset address of ________ segment.
   A. Data segment [✓ B. Code segment] C. Stack segment D. Extra segment
5. If offset is a 13-bit number address, then the maximum possible locations are_____.
   A. 4 KB [✓ B. 8 KB] C. 64 KB D. 16KB
6. The equivalent of instruction XOR BL,FFH is :
   A. NEG BX [✓ B. NOT BL] C.AND AL,0 D. OR AL,FF
7. The content of register BL after execution the instruction MOV BL,-25:
   A. BL=E5H B. BL=F2H [✓ C. BL=E7H] D. BL=F5H
8. In the following instruction sequence, show the value of AL, in binary:

```
MOV AL,5
NEG AL
XOR AL,FFH
```

   A. AL= 11111011B [✓ C. AL= 00000100B]
   B AL= 11110101B D. AL= 01011111B
9. Find the content of AL and zero flag (ZF) after execution of the following code:

```
MOV AL,58H
TEST AL, 4
```

   [illegible] AL=0 ; ZF=1 B.AL=58H ; ZF=0 [✓ C. AL=58H; ZF=1] D.AL=0 ; ZF=0
10. If the SP register contains the hexadecimal value F200H, what will be its value after executing the following program segment?

```
PUSH AX
PUSH BX
XCHG AH,BH
POP AX
POP BX
```

   A. F202H B. F208H C. F204H [✓ D. F200H]

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Question #2 (12 points)

A.
```
ARR DW 3333H, 4444H, 1111H,
2222H, 0FFFH
      MOV BX,OFFSET ARR
      MOV SI, 6

      MOV DI,2

      MOV DL, [BX ][SI]
      MOV AL, [BX][ DI]
      XCHG AL, DL
```

AL= D4pC(1) ; DL= 44H

B.
```
MOV AL, 22H
   AND AL, 03H
   JZ YES
   NOT AL
   JMP EE
   YES: NEG AL
   EE:HLT
```

AL= ~~EE~~ FD

7

C.
```
ARRAY   DB  8, 3, 2, 4,6,8,5,7,0
     MOV AL,0
     MOV CX,4
     MOV BX, OFFSET ARRAY
     L:   STC
     ADC AL, [BX]
     ADD BX,2
     LOOP L
    MOV   AH,[BX]
```

AL= 19 H   AH= 00 H

D.
```
X DW 104FH, 201AH, C030H
     MOV BX, OFFSET  X
     MOV AX, [BX]
     INC BX
     MOV   CX,[BX]
     PUSH AX
     PUSH CX
     POP AX
     POP CX
```

AX= 1A10 H   CX= 104F H

E.
```
 MOV BL,37H
MOV CX,7
MOV AX,0
MOV DL,1
L: ADD AX,CX
SHL DL,1
TEST BL,DL
LOOPNZ L
```

*

CX=CX-1
CX≠0 ZF=0

AX= ~~[illegible]~~ 20H ; CX=......~~7~~ 00 H

DL=... ~~[illegible]~~ 00 H
~~20H~~

F.
```
MOV AX, 0004H
 MOV CL, 4H
 XOR BH, BH
 STC
 RCL BH, CL
 MUL BH
```

BH= ~~[illegible]~~ 08 H

AX= ~~20H~~
24 D

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Question #3 (8 points)

a) Suppose we have a microprocessor with 32bits address bus; what is the maximum memory capacity will be?

$2^{32} = 2^{2} \times 2^{30} = 4GB$ ✓ 1

b) For the following instruction MOV [BX], 50H; what is the addressing mode used?

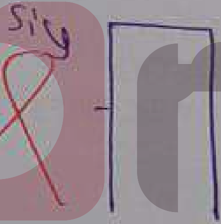
Base relative addressing mode 1

c) Write an assembler codes to clear 1KB of memory locations defined as byte.

```
X DB 1024, 20, 40.....
```

$2^{10} = 1024$ Byte

```
    MOV CX, 1024
Q:  MOV SI, offset X
    mov BL, [SI]
    AND BL, 00H
    inc si
    dec cx
CMP CX, 1024
JL Q
H It: Exit
```



2

d) Write the equivalent code of the following instructions

- LOOP L

```
Dec cx
JNZ L1
```

✓ 1

- SUB AL,BL

```
XOR AL, BL
```

1

- XOR AX,FFFFH

```
NOT AX
```

✓ 1

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